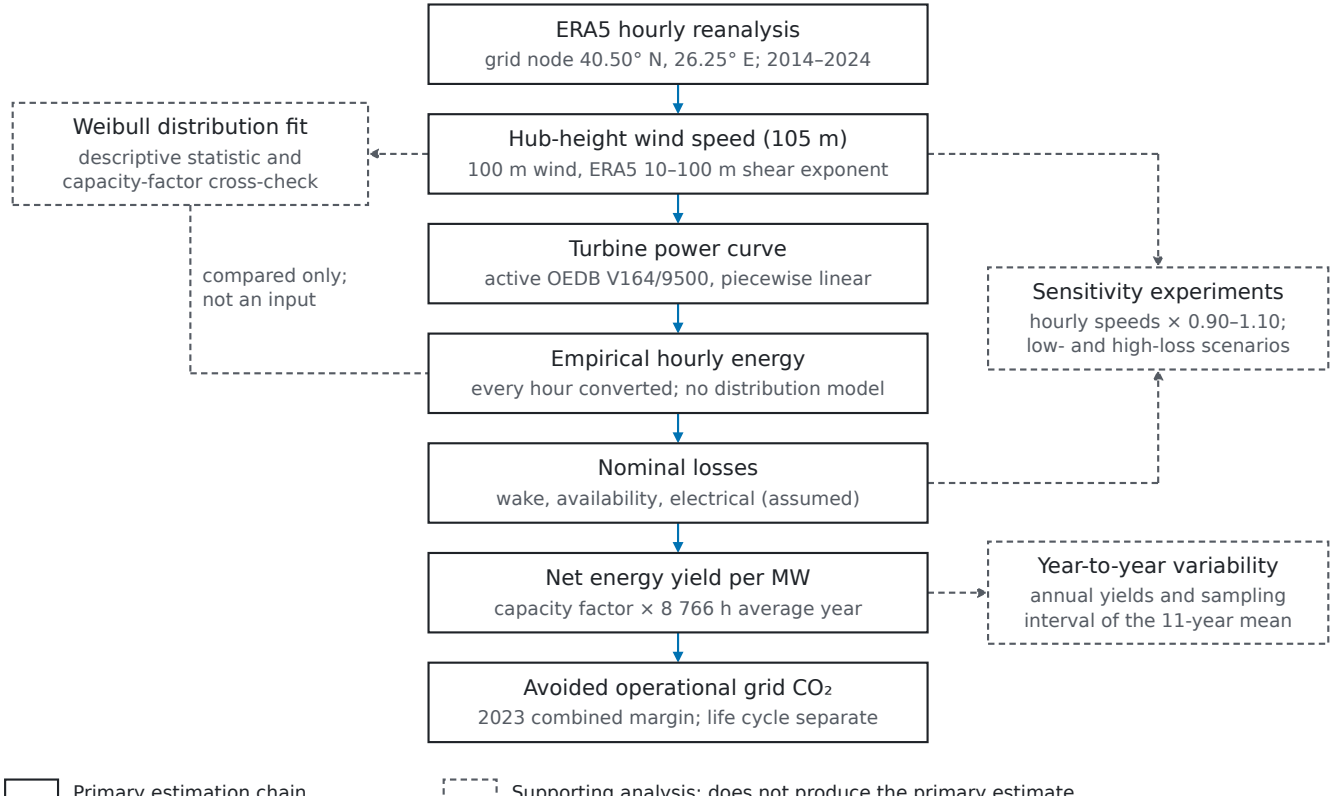
ERA5 hourly reanalysis
grid node 40.50° N, 26.25° E; 2014–2024
Hub-height wind speed (105 m)
100 m wind, ERA5 10–100 m shear exponent
Turbine power curve
active OEDB V164/9500, piecewise linear
Empirical hourly energy
every hour converted; no distribution model
Nominal losses
wake, availability, electrical (assumed)
Net energy yield per MW
capacity factor × 8 766 h average year
Avoided operational grid CO₂
2023 combined margin; life cycle separate
Weibull distribution fit
descriptive statistic and capacity-factor cross-check
compared only; not an input
Sensitivity experiments
hourly speeds × 0.90–1.10; low- and high-loss scenarios
Year-to-year variability
annual yields and sampling interval of the 11-year mean
Primary estimation chain
Supporting analysis; does not produce the primary estimate